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Marks	0/20
Grade	0 out of 100

Question 1

Not answered

Marked out of 1

A researcher is conducting a literature search to write a grant application. He comes across two RCTs evaluating same treatment against placebo. Trial A shows a difference in favour of treatment with a p value 0.03; Trial B shows a difference with p-value reaching 0.09 when using a significantly larger sample size. Which of the following explanations is plausible?

Select one:

- ☐ Trial A shows statistically significant difference
- ☐ Trial A is more representative of true difference
- ☐ P value reflects size of treatment effect
- ☐ Smaller samples always generate smaller p values.
- ☐ Dose of administered medication cannot influence treatment effect

Your answer is incorrect.

Smaller samples may generate larger p-values obscuring a true association that may be present. P-values do not reflect the size of treatment effect - they only signify the level of statistical probability attached to an estimate. Clinical variables such as patient profiles and the dose of medications could affect the effect size. Arbitrarily, p-value less than 0.05 is taken as a measure of statistical significance.

The correct answer is: Trial A shows statistically significant difference

Question 2

Not answered

Marked out of 1

If we change the confidence level from 95% to 90% when constructing a confidence interval for a population mean, we can expect the size of the interval to

Select one:

- ☐ Increase
- ☐ Increase in positive direction only
- ☐ Increase in negative direction only
- ☐ Decrease
- ☐ Not change

Your answer is incorrect.

If the degree of confidence is raised, the intervals or range of values within which we can call the true mean value to be present will also increase. At times, this may even cross point of no significance. Arbitrarily, similar to the p-value of 0.05, we use confidence limits of 95%.

The correct answer is: Increase



Question 3

Not answered

Marked out of 1

If the 95% confidence limits for the average haloperidol dose used during rapid tranquillisation through IM route are 7.5 and 12.5, which of the following could be the 90% confidence limits?

Select one:

- ☐ 6 and 11 mg
- ☐ 8 and 11 mg
- ☐ 9 and 15 mg
- ☐ 7.5 and 12.5 mg
- ☐ 9 and 12.5 mg

Your answer is incorrect.

As the degree of confidence has been reduced, the range within which we can call the true value to be present can be narrowed further (making it appear more precise). Hence, 8 to 11mg (range of) seems more appropriate than other choices.

The correct answer is: 8 and 11 mg



Question 4

Not answered

Marked out of 1

A 95% confidence interval for a population mean can be inferred as

Select one:

- ☐ 95% of the population has been sampled
- ☐ Having 95% confidence that the population mean will be the same as the point estimate of sample mean
- ☐ 95% of the population will have the parameter value between the limits
- ☐ Having 95% confidence that the population mean will fall within the interval.
- ☐ Having 95% confidence that the sample mean will remain as the same point estimate if repeated samplings are done in the population.

Your answer is incorrect.

95% confidence interval for an estimate (such as mean) can be interpreted as the range of values within which we are 95% confident that the true population estimate (mean) lies. All other interpretations given above are inaccurate.

The correct answer is: Having 95% confidence that the population mean will fall within the interval.



Question 5

Not answered

Marked out of 1

Which of the following measures increases the power of a study?

Select one:

- ☐ Including heterogenous patient groups
- ☐ Using more than 2 primary outcome measures
- ☐ Comparing two active treatments
- ☐ Longer period of follow-up in a trial
- ☐ Comparing active treatment vs. placebo

Your answer is incorrect.

If active treatment and placebo are compared, the effect may be bigger resulting in a powerful study.

The correct answer is: Comparing active treatment vs. placebo



Question 6

Not answered

Marked out of 1

Which of the following terms describe a type 1 error?

Select one:

- ☐ False negative
- ☐ False positive
- ☐ True negative
- ☐ True positive
- ☐ Total negative

Your answer is incorrect.

Rejecting the null hypothesis when it is, in fact, true is same as claiming a false positive result. This is called type 1 error.

The correct answer is: False positive



Question 7

Not answered

Marked out of 1

The most important difference between inferential and descriptive statistical analyses is that

Select one:

- ☐ The objective of descriptive statistics is to communicate the results without attempting to generalize
- ☐ Descriptive analysis aids in generalisation of the results
- ☐ Inferential analysis is used more commonly in cross-sectional surveys
- ☐ Inferential analysis cannot be carried out with a smaller data-set
- ☐ Descriptive analysis is based on hypothetical assumptions

Your answer is incorrect.

Descriptive statistics helps to organize and summarize data in easily communicable manner. It does not attempt to make inferences. It is not based on prior assumptions, and no hypothesis testing takes place.

The correct answer is: The objective of descriptive statistics is to communicate the results without attempting to generalize



Question 8

Not answered

Marked out of 1

A randomized trial comparing the efficacy of two drugs showed a difference between the two (with a P value < .05). Assume that in reality the two drugs do not differ. This refers to

Select one:

- ☐ 1 - α
- ☐ Random error
- ☐ Type II error (β error)
- ☐ 1 - β
- ☐ Type I error (alpha error)

Your answer is incorrect.

A type I error is committed by not accepting the null hypothesis when, in fact, it is true. The probability (p) of making a type I error is designated by the symbol [α]. The significance level is the value for which α is set at for testing the hypothesis (usually 0.05). It is an arbitrary value chosen as the cut-off point for which the p-value is considered significant or not significant. 'Chance findings' written up as conclusive reports constitute type 1 error.

The correct answer is: Type I error (alpha error)

Question 9

Not answered

Marked out of 1

A study examining the association between in-utero exposure to famine and schizophrenia uses a one-sided significance test. Which of the following underlying assumptions justify the one-sided rather than a two-sided test

Select one:

- ☐ The expected pattern of association is one of a linear relationship
- ☐ The in utero exposure to famine among schizophrenia patients is normally distributed
- ☐ It is expected that in utero famine exposure causes schizophrenia
- ☐ The likelihood of type 2 error is expected to be high with two sided test
- ☐ Only one control group was used

Your answer is incorrect.

Clinical research is generally carried out to determine the relationship between two variables - intervention and treatment effect; risk factor and prognosis, exposure and disease, etc. Such relationships can be present in only one direction (e.g., an increase, a decrease, an improvement, or harm) or any direction. In the former case, a one-tailed hypothesis is employed while, in the latter case, we use two-tailed hypotheses. An example of a two-tailed test is the investigation of a new drug, which may be helpful or harmful. As it seems highly likely that famine exposure causes schizophrenia rather than the vice versa, one-tailed hypothesis is justified here. The major advantage of a one-tailed test is that a smaller number of study subjects are needed to reach a certain power, as compared to a two-tailed test.

The correct answer is: It is expected that in utero famine exposure causes schizophrenia

Question 10

Not answered

Marked out of 1

A new drug is being tested to treat akathisia induced by neuroleptics. All of the following measures will improve the statistical power of this RCT except

Select one:

- ☐ Reducing drop-out rates by monitoring adherence
- ☐ Selecting more homogenous clinical sample
- ☐ Enrolling more patients in the study
- ☐ Requiring a significantly high difference in proportion of responders between placebo and active arms
- ☐ Conducting subgroup analysis on patients with a shorter duration of illness

Your answer is incorrect.

The power of a study to detect a difference that exists in the population increases with increasing sample size. It reduces with increasing variance in the sample. Power is $(1 - \beta)$; it reduces as the significance level is made smaller (i.e., alpha or p is set too low) provided other factors are kept constant. Power increases as the absolute value of the distance between the null and alternative means increases (effect size).

The correct answer is: Requiring a significantly high difference in proportion of responders between placebo and active arms

Question 11

Not answered

Marked out of 1

As the sample size increases, the confidence interval for the population mean will

Select one:

- ☐ Increase in negative direction only
- ☐ Not change
- ☐ Increase
- ☐ Decrease
- ☐ Increase in positive direction only

Your answer is incorrect.

Precision increases with sample size; hence the confidence interval will decrease for a given degree of confidence.

The correct answer is: Decrease



Question 12

Not answered

Marked out of 1

Power calculation is best done at which of the following stages of a randomized controlled trial

Select one:

- ☐ When writing up trial report to a journal
- ☐ Following randomization
- ☐ When planning a trial
- ☐ Following trial completion
- ☐ Following treatment completion

Your answer is incorrect.

The power calculation is best done when planning a trial - post hoc calculations are weak in principle and not advisable.

The correct answer is: When planning a trial



Question 13

Not answered

Marked out of 1

In a trial of a novel peripheral acting drug for symptoms of anticipatory anxiety in panic disorder/agoraphobia the sample size required was calculated to be 300 patients for RCT. Unfortunately, due to managerial changes in the trial administration board the trial could include only 175 patients. Which if any of the following statements is true

Select one:

- ☐ The chance of both errors is increased.
- ☐ The chance of a type 2 error is reduced.
- ☐ The chance of a type 2 error may increase.
- ☐ The chance of a type 1 error may increase.
- ☐ The chance of both errors is unaltered.

Your answer is incorrect.

Imagine that you are proposing that a species of blue rabbit exists in England. Suppose the truth is that there are no blue rabbits. Type 1 error refers to the situation where you get hold of a dirty rabbit and claim that it is blue; i.e. a 'false positive result'. The chance of type 1 error is not directly dependent on sample size. You can examine just three rabbits in a bush and claim one is blue. In contrast type 2 error refers to not finding an effect that is present. Suppose the truth is there are few blue rabbits in Shropshire. This is possible if one does not examine enough specimens of rabbits, one may conclude that a blue rabbit does not exist i.e. a 'false negative result'. So type 2 error depends on sample size. BMJ 2008;337:a2957

The correct answer is: The chance of a type 2 error may increase.

Question 14

Not answered

Marked out of 1

In a study on degree of recovery achieved following an episode of psychotic depression, the study group contacts multiple centres in a bid to increase the size of its random sample from 100 to 900. Which of the following can be observed?

Select one:

- ☐ Have no effect because the population size is the same
- ☐ Reduction in systematic errors
- ☐ Reduction in standard deviation of the observations
- ☐ Reduction in precision due to data contamination
- ☐ Reduction in the standard error of the estimate

Your answer is incorrect.

When the sample size is increased, precision will increase, the standard error will reduce. There will be no effect on a bias (systematic error) and mostly, no effect on standard deviation.

The correct answer is: Reduction in the standard error of the estimate

Question 15

Not answered

Marked out of 1

With respect to inferential statistics, which of the following is not correct?

Select one:

- ☐ The probability of a type I error is related to alpha
- ☐ Two tailed hypothesis may have different significance levels than one tailed hypothesis for the same research question
- ☐ The probability of type II error depends on the sample size
- ☐ The p value indicates the probability of null hypothesis being false
- ☐ The power of a test depends on the sample size

Your answer is incorrect.

The p-value indicates the probability that the null hypothesis is true. The probability of a type 1 error is related to alpha (gives the p-value). The probability of type 2 error is related to beta, power of a study; it depends on sample size. Two-tailed hypothesis testing may need adjustment in p-value (generally halved) compared to one-tailed test with same data.

The correct answer is: The p value indicates the probability of null hypothesis being false



Question **16**

Not answered

Marked out of 1

In large trials where multiple resources are used, investigating teams may use the final dataset to answer various different questions they might have. Such multiple testing leads to

Select one:

- ☐ Type 1 error
- ☐ Type 2 error
- ☐ Selection bias
- ☐ Type 3 error
- ☐ Confounding error

Your answer is incorrect.

Repeated testing of hypothesis using a same data, multiple subset analysis and secondary analysis can lead to type 1 error as in these cases at least one test will be positive out of 20 if 'p' is set at 0.05.

The correct answer is: Type 1 error



Question 17

Not answered

Marked out of 1

From the following options choose the correct pair of statistical measure and value of no difference with respect to confidence intervals:

Select one:

- ☐ Relative risk reduction : zero
- ☐ Relative risk : zero
- ☐ Absolute risk reduction : zero
- ☐ Odds ratio : infinity
- ☐ Number needed to treat : one

Your answer is incorrect.

With respect to confidence intervals of ratios, 'one' is the value of no difference. For confidence interval of differences in mean, it is 'zero'. For inverse ratios such as NNT, it is infinity.

The correct answer is: Absolute risk reduction : zero



Question 18

Not answered

Marked out of 1

In a study exploring the presence of suicidal ideation in those psychiatric patients who are currently smoking compared to non-smokers, the following was found: odds ratio=2.00, 95% confidence interval=1.48-2.70, $p < 0.0001$. Which of the following can be inferred from the above?

Select one:

- ☐ The confidence interval does not show the size of the effect
- ☐ Degree of bias can be estimated from reported confidence interval
- ☐ The risk that given odds ratio 2.00 is a chance observation is 5%
- ☐ The confidence interval embraces value of no difference
- ☐ None of the listed inferences are possible

Your answer is incorrect.

Here the confidence interval does not cross 1 - so it does not embrace the value of no difference. The confidence interval shows the size of the effect - nearly two times. One cannot deduct the degree of bias from reported confidence interval. The risk that given odds ratio 2 is due to chance is 1 in 10000 (from p value).

The correct answer is: None of the listed inferences are possible

Question 19

Not answered

Marked out of 1

A study sets out to estimate the difference in average waiting time between CBT and IPT. Which of the following values, if crossed by the 95% confidence interval of the final point estimate, will make the findings non-significant?

Select one:

- ☐ 95
- ☐ 0
- ☐ 1
- ☐ 1.96
- ☐ Infinity

Your answer is incorrect.

For a difference in mean values, if the 95% CI crosses 0, then it loses its significance.

The correct answer is: 0



Question 20

Not answered

Marked out of 1

Calculate the standard error of mean weight gained by a sample of patients on Olanzapine if 25 were included in a study that yielded a standard deviation of 10kg for the period observed:

Select one:

- ☐ 250
- ☐ 2.5
- ☐ 5
- ☐ 0.4
- ☐ 2

Your answer is incorrect.

$SE = SD / \sqrt{n} = 10 / \sqrt{25} = 10 / 5 = 2$

The correct answer is: 2



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